

TOULON HYERES AP, France

WMO: 076670

Lat: 43.0944N

Lon: 6.1464E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.9	35.3	14.1	11.2	40.5	18.1	13.7	42.8	29.3	53.3	27.0	54.0	6.4	330	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	87.8	69.8	85.6	70.2	83.8	70.1	75.8	82.8	74.3	81.6	72.9	80.4	12.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	124.4	81.1	71.8	117.3	79.7	70.2	111.0	78.5	39.2	83.1	37.8	81.4	36.5	80.6	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	23.2	20.6	DB	28.9	94.2	2.5	2.5	27.1	96.0	25.7	97.4	24.3	98.8	22.5	100.7	(4)
(5)			WB	26.2	77.3	2.5	1.5	24.3	78.3	22.9	79.2	21.5	80.0	19.6	81.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	48.3	48.3	52.1	56.1	62.5	69.2	74.2	74.1	68.4	62.6	54.8	50.0	(6)	
	DBStd	10.35	4.70	4.65	4.40	3.99	4.25	4.59	3.86	3.80	4.18	4.44	5.18	4.92	(7)	
	HDD50	273	88	78	25	4	0	0	0	0	0	0	17	61	(8)	
	HDD65	2653	517	469	399	267	100	14	1	1	15	97	307	466	(9)	
	CDD50	3972	36	30	92	187	389	577	750	748	552	391	160	60	(10)	
	CDD65	877	0	0	1	0	24	141	286	285	117	23	0	0	(11)	
	CDH74	7053	0	0	2	3	118	1006	2594	2574	701	55	0	0	(12)	
	CDH80	1652	0	0	0	0	13	181	686	684	86	2	0	0	(13)	
(14) Wind	WSAvg	8.5	8.8	8.9	9.4	9.3	8.5	8.0	8.5	7.8	8.0	8.0	8.7	8.6	(14)	
(15) Precipitation	PrecAvg	28.3	3.0	2.1	1.8	2.5	1.7	1.4	0.4	0.7	3.0	4.2	4.4	3.0	(15)	
	PrecMax	45.5	8.8	5.9	5.9	6.3	4.0	5.5	1.8	3.1	7.7	10.0	13.0	8.4	(16)	
	PrecMin	15.2	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.2	1.1	0.9	0.2	(17)	
	PrecStd	7.5	2.6	1.8	1.5	1.7	1.2	1.4	0.5	0.7	2.0	2.2	2.9	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	64.4	70.8	73.2	81.4	88.1	91.7	91.3	85.4	78.1	69.8	65.5	(19)	
		MCWB	54.0	52.2	55.6	58.0	63.8	69.2	67.7	69.7	67.0	64.6	60.1	55.1	(20)	
	2%	DB	61.3	61.1	65.6	70.1	77.0	84.3	87.8	87.9	82.3	75.3	67.3	62.3	(21)	
		MCWB	51.8	51.1	53.7	57.7	62.5	68.9	69.4	70.4	67.4	64.8	59.6	54.5	(22)	
	5%	DB	59.0	59.1	63.0	67.5	74.2	81.8	85.6	85.7	80.2	73.3	65.7	60.5	(23)	
		MCWB	51.9	51.2	53.1	56.9	62.1	68.7	70.5	70.6	67.5	64.4	58.8	54.2	(24)	
	10%	DB	57.2	57.4	61.1	65.1	71.9	79.5	83.8	83.8	78.1	71.4	64.1	58.8	(25)	
		MCWB	51.2	50.5	52.8	55.9	61.2	67.9	70.2	70.7	66.7	63.4	57.9	53.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	55.7	58.2	62.1	68.3	75.4	77.4	78.3	73.6	69.7	63.9	58.8	(27)	
		MCDB	60.8	59.6	64.9	68.1	75.1	82.1	84.2	84.6	79.4	74.0	67.2	61.3	(28)	
	2%	WB	55.1	54.3	56.6	60.2	66.1	73.0	75.6	76.3	71.8	68.0	62.0	57.1	(29)	
		MCDB	58.8	58.3	62.2	66.4	73.3	80.3	82.8	83.0	78.5	72.5	65.3	60.3	(30)	
	5%	WB	53.6	53.0	55.4	58.8	64.5	71.3	74.2	74.6	70.3	66.5	60.6	55.7	(31)	
		MCDB	57.6	57.4	60.7	64.8	71.3	78.7	81.7	81.8	77.1	71.2	64.1	59.2	(32)	
	10%	WB	52.0	51.6	54.2	57.5	63.1	69.7	72.8	73.2	68.7	65.0	59.1	54.0	(33)	
		MCDB	56.3	56.5	59.5	63.1	69.6	77.2	80.6	80.7	75.5	69.9	63.2	57.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	16.0	16.5	16.0	16.4	17.6	18.0	18.5	17.3	15.8	15.0	14.8	(35)	
		MCDBR	17.0	18.2	18.9	19.3	19.6	19.8	20.2	20.7	19.5	17.4	16.3	16.3	(36)	
		MCWBR	12.1	13.0	12.2	11.4	11.0	10.7	10.8	11.7	11.5	10.8	11.2	11.5	(37)	
	5% WB	MCDBR	13.8	16.2	17.3	17.7	17.8	18.0	18.1	18.7	17.3	14.8	13.3	13.6	(38)	
		MCWBR	11.4	12.8	12.3	11.9	11.4	11.1	11.6	12.6	11.7	10.4	10.4	10.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.331	0.349	0.385	0.407	0.417	0.416	0.410	0.406	0.398	0.392	0.357	0.327	(40)	
	taud		2.476	2.424	2.340	2.296	2.313	2.351	2.371	2.389	2.397	2.399	2.443	2.490	(41)	
	Ebn at Noon		254	268	271	272	272	272	270	263	248	240	240	244	(42)	
	Edh at Noon		23	29	35	40	40	39	38	36	33	30	24	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		586	872	1312	1700	2039	2352	2351	2046	1544	997	639	508	(44)	
	RadStd		56	87	108	110	150	91	92	69	81	79	65	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79

Nomenclature: See separate page